

11 (**long**) A number of large display boards are required for a conference on the sustainability of materials. Each board consists of a rectangular panel of material whose length L and width b are fixed, and whose thickness d may vary. The panels are most heavily loaded by their self-weight when being carried horizontally, simply supported along their edges of width b . As the boards are to be used as a design case study, the objective is to minimise their environmental impact.

(a) Show that the central deflection δ and the maximum bending stress σ_{max} in the panel, due to its self-weight, are given respectively by

$$\delta = \frac{5\rho g L^4}{32Ed^2} \quad \text{and} \quad \sigma_{max} = \frac{3\rho g L^2}{4d}$$

where ρ is the density, E is Young's modulus, and g is the acceleration due to gravity. [8]

(b) The length and width of one set of panels are 3 m and 2 m respectively, and the thickness must be between 5 and 20 mm. For each of the materials in Table 1, find the minimum thickness required to meet both a stiffness constraint that the maximum allowable deflection is 20 cm, and a strength constraint that the maximum stress must not exceed half the failure strength, σ_f . Eliminate any materials that fail to meet the target range of thickness. [10]

(c) The organisers have imposed upper limits of £30 per panel and 35 kg per panel. Eliminate any of the materials identified in part (b) that fail to meet either constraint. For the remaining shortlist, evaluate the embodied energies, and identify the material with the lowest value. [7]

(d) All the materials are imported, and travel a distance of 1000 km. Evaluate the transportation energies for truck freight, for which the energy consumption is $0.94 \text{ MJ tonne}^{-1} \text{ km}^{-1}$. Does the ranking of materials change, based on the total embodied and transportation energy? [5]

Material	Density, ρ (kg m ⁻³)	Young's modulus, E (GPa)	Failure strength, σ_f (MPa)	Cost / kg, C_m (£ kg ⁻¹)	Embodied energy / kg, C_e (MJ kg ⁻¹)
Al foam	360	0.9	1.3	9	370
Biocomposite	530	3	10	0.8	7
Pine	600	10	40	0.8	12
Rigid polymer foam	320	0.3	6	8	81
Fibreboard	750	4	10	0.3	18

Table 1